

## Practical-4

**Aim:** Write a program to input the line coordinates from the user to generate a line using Bresenham's Algorithm.

**Program:**

```
#include <iostream>
#include <graphics.h>
using namespace std;
int main() {
    int gd = DETECT, gm;
    initgraph(&gd, &gm, (char*)"");
    int x1, y1, x2, y2;
    x1=50,x2=500,y1=50,y2=500;
    // Draw text (name)
    outtextxy(150, 50, "Name: Gagan kumar shah, URN:2302716");
    int dx = x2 - x1;
    int dy = y2 - y1;
    int x = x1;
    int y = y1;
    int p = 2 * dy - dx;
    putpixel(x, y, WHITE);
    while (x < x2) {
        x++;
        if (p < 0) {
            p += 2 * dy;
        } else {
            y++;
            p += 2 * (dy - dx);
        }
        putpixel(x, y, WHITE);
    }
}
```

```
    getch();  
    closegraph();  
    return 0;  
}
```

### Output:

